

MATH 2630-090: TEST 1 – Summer 2024

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Name (*please print*): _____

Instructions: Please read the following instructions carefully.

- **Read all of the directions** for each problem.
- You have **75 minutes** to finish the exam.
- **Show work to receive credit.** Answers that are wrong, but still have work might receive partial credit. If you provide the answer to a computation without work, I will not give partial credit.
- Provide **exact** solutions.
- Do **not** write in the table below.

Problem	Points	Score
1	10	
2	12	
3	10	
4	15	
5	10	
6	12	
7	15	
8	15	
Total	99 +1	

1. (8 points) Find the domain of the vector function.

$$\vec{r}(t) = \left\langle \frac{t}{\sqrt{t-3}}, \frac{1}{(t+6)(t-5)}, \ln(t+1) \right\rangle$$

2. (12 points, 3 each) Use the vectors $\vec{v} = 4\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{w} = -2\hat{i} - \hat{j} + 2\hat{k}$ to answer the following.

- (a) $\vec{v} \cdot \vec{w}$
- (b) $|\vec{v} - \vec{w}|$
- (c) A *unit* vector orthogonal to both $\vec{v}$ and $\vec{w}$.
- (d) The angle θ between $\vec{v}$ and $\vec{w}$ (Round to 2 decimals).

3. (10 points) Find the parametric equation for a line which passes through points $P(1, -1, 3)$ and $Q(-2, 0, 7)$.

4. (15 points, 3,7,5 resp.) Use the two lines L_1 and L_2 to answer the following.

$$L_1 : \frac{x_1 - 3}{-2} = \frac{y_1 + 1}{-4} = \frac{z_1}{2} \quad L_2 : x_2 = 3 - 6t_2, y_2 = 1 + 2t_2, z_2 = -1 - 4t_2.$$

- (a) Are L_1 and L_2 parallel? Justify your answer.
- (b) Are L_1 and L_2 skew lines or do they intersect? If they intersect, at which point?
- (c) Where does L_1 cross the xz -plane?

5. (10 points) Find the equation for the plane which contains points $P(-5, 2, 0)$ and $Q(1, 1, 1)$ and is perpendicular to the plane $x - y + 2z = 5$.

6. (12 points) Evaluate the limit of the vector function, $\lim_{t \rightarrow 1} \vec{r}(t)$.

$$\vec{r}(t) = (te^{-t})\hat{i} + \left(\frac{5t^2 - 5}{t - 1}\right)\hat{j} + (\cos(t - 1) - \sin(t - 1))\hat{k}.$$

7. (15 points) Find the equation of the tangent line to the space curve given by the vector function $\vec{r}(t)$ at the time $t = 4$.

$$\vec{r}(t) = \langle \sqrt{t}, t + 2, t^2 \rangle.$$

8. (15 points) If $\vec{r}'(t) = \sin(t)\hat{i} + t\hat{j} + \hat{k}$ and $\vec{r}(0) = -\hat{i} + 3\hat{k}$. Find $\vec{r}(t)$.